

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 15: Similarity (With Applications to Maps and Models)

Previous year board questions, topic-wise: 2 questions

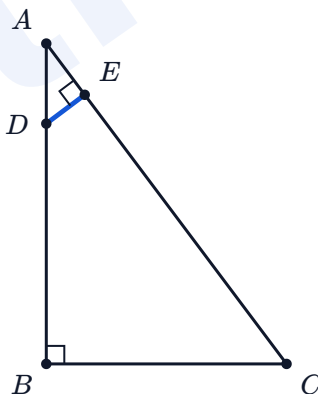
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Similar triangles

1.  $\triangle ABC \sim \triangle EFG$  with  $\angle ABC = \angle EFG = 60^\circ$ ,  $AB = 15$  cm,  $BC = 3$  cm and  $EF = 75$  cm. Find  $FG$ . (a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm **ICSE 2025, 1 mark**

## 2 Maps and models

2. In  $\triangle ABC$ ,  $\angle ABC = 90^\circ$ ,  $AB = 20$  cm and  $AC = 25$  cm.  $D$  is a point on  $AB$  and  $E$  is a point on  $AC$  with  $DE \perp AC$  and  $DE = 3$  cm. (a) Prove that  $\triangle ABC \sim \triangle AED$ . (b) Find  $BC$ ,  $AD$  and  $AE$ . (c)  $BCED$  is a plot of land drawn on a map, and its actual area is  $576$  m<sup>2</sup>. Find the scale factor of the map. **ICSE 2025, 4 marks**



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